



Top 121 CAT Arrangement Questions With Video Solutions

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Questions

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five women decided to go shopping to M.G. Road, Bangalore. They arrived at the designated meeting place in the following order:

1. Archana,
2. Chellamma,
3. Dhenuka,
4. Helen, and
5. Shahnaz.

Each woman spent at least Rs. 1000. Below are some additional facts about how much they spent during their shopping spree.

- i. The woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193.
- ii. One woman spent Rs. 1340 and she was not Dhenuka.
- iii. One woman spent Rs. 1378 more than Chellamma.
- iv. One woman spent Rs. 2517 and she was not Archana.
- v. Helen spent more than Dhenuka.
- vi. Shahnaz spent the largest amount and Chellamma the smallest.

Question 1

Which of the following amounts was spent by one of them?

- A Rs. 1139
- B Rs. 1378
- C Rs. 2571
- D Rs. 2718

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Question 2

What was the amount spent by Helen?

- A Rs. 1193
- B Rs. 1340
- C Rs. 2234
- D Rs. 2517

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Question 3

The woman who spent Rs. 1193 is

- A Archana
- B Chellamma
- C Dhenuka
- D Helen

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Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Seven varsity basketball players (A, B, C, D, E, F, and G) are to be honoured at a special luncheon. The players will be seated on the dais in a row. A and G have to leave the luncheon early and so must be seated at the extreme right. B will receive the most valuable player's trophy and so must be in the centre to facilitate presentation. C and D are bitter rivals and therefore must be seated as far apart as possible.

Question 4

Which of the following pairs cannot occupy the seats on either side of B?

- A F & D
- B D & E
- C E & G
- D C & F

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Question 5

Which of the following cannot be seated at either end?

- A C
- B D
- C F
- D G

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Question 6

Which of the following pairs cannot be seated together?

- A B & D
- B C & F

C D & G

D E & A

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Instructions

DIRECTIONS for the following two questions: Answer the questions on the basis of the information given below.

The Head of a newly formed government desires to appoint five of the six elected members A, B, C, D, E and F to portfolios of Home, Power, Defence, Telecom and Finance. F does not want any portfolio if D gets one of the five. C wants either Home or Finance or no portfolio. B says that if D gets either Power or Telecom then she must get the other one. E insists on a portfolio if A gets one.

Question 7

If A gets Home and C gets Finance, then which is NOT a valid assignment of Defense and Telecom?

- A D-Defence, B-Telecom.
- B F-Defence, B-Telecom.
- C B-Defence, E-Telecom.
- D B-Defence, D-Telecom.

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Question 8

Which is a valid assignment?

- A A-Home, B-Power, C-Defence, D-Telecom, E-Finance.
- B C-Home, D-Power, A-Defence, B-Telecom, E-Finance.
- C A-Home, B-Power, E-Defence, D-Telecom, F-Finance.
- D B-Home, F-Power, E-Defence, C-Telecom, A-Finance.

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Instructions

Answer the questions on the basis of the information given below.

A, B, C, D, E, and F are a group of friends. There are two housewives, one professor, one engineer, one accountant and one lawyer in the group. There are only two married couples in the group. The lawyer is married to D, who is a housewife. No woman in the group is either an engineer or an accountant. C, the accountant, is married to F, who is a professor. A is married to a housewife. E is not a housewife.

Question 9

Which of the following is one of the married couples?

- A A & B

- B B & E
- C D & E
- D A & D

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Question 10

How many members of the group are males?

- A 2
- B 3
- C 4
- D Cannot be determined

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Question 11

What is E's profession?

- A Engineer
- B Lawyer
- C Professor
- D Accountant

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Instructions

The pages of a book are numbered 0, 1, 2 . upto M, $M > 0$. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are :

1. OPEN : Position the book at page No. 1
2. CLOSE : Position the book at page No. 0
3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M.
4. BACKWARD, n : From the current page, move backward by n pages; if in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

Question 12

FORWARD, 5; BACKWARD, 4. Which of the following statements about the above instructions is true?

- A $n_2 = n_1 + 4$ provided $1 < n_1 < 7$
- B $n_2 = n_1$ provided $M < 6$
- C $n_2 = n_1 + 1$ provided $M - n_1 > 5$
- D $n_2 - n_1 < 0$ provided $M > 0$



VIDEO SOLUTION



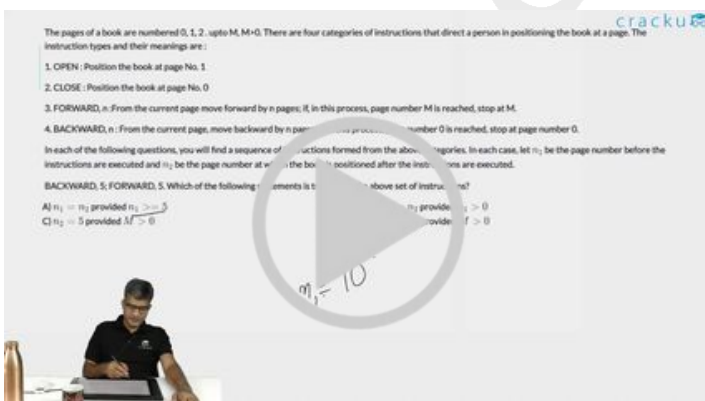
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Question 13

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

- A $n_1 = n_2$ provided $n_1 \geq 5$
- B $n_1 = n_2$ provided $n_1 > 0$
- C $n_2 = 5$ provided $M > 0$
- D $n_1 > n_2$ provided $M > 0$



VIDEO SOLUTION

Question 14

FORWARD, 25 ; BACKWARD, 10. which of the following statements is true?

- A $n_1 = n_2$ if $M=10$ and $n_1 = 0$
- B $M = 20$ provided $n_1 > 0$
- C $n_1 > 30$ provided $M = 900$
- D $n_1 = 37$ provided $M = 25$

The pages of a book are numbered 0, 1, 2, ... upto M, $M > 0$. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are :

1. OPEN : Position the book at page No. 1
2. CLOSE : Position the book at page No. 0
3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M .
4. BACKWARD, n : From the current page, move backward by n pages; if, in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

A) $n_1 = n_2$ provided $n_1 > 5$ B) $n_1 = n_2$ provided $n_1 > 0$
C) $n_2 = 5$ provided $M > 0$ D) $n_2 = 5$ provided $M > 5$

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Question 15

FORWARD, 10; FORWARD, 10. Which of the following statements about the above instructions is true?

- A $n_2 - n_1 = 20$ only if $n_1 = 0$
- B $n_2 - n_1 = 20$ if $M > 20$ and $n_1 = 1$
- C $n_2 - n_1 = 10$ if $M = 21$ and $n_1 = 0$
- D $n_2 > n_1$ if $M > 0$

The pages of a book are numbered 0, 1, 2, ... upto M, $M > 0$. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are :

1. OPEN : Position the book at page No. 1
2. CLOSE : Position the book at page No. 0
3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M .
4. BACKWARD, n : From the current page, move backward by n pages; if, in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

A) $n_1 = n_2$ provided $n_1 > 5$ B) $n_1 = n_2$ provided $n_1 > 0$
C) $n_2 = 5$ provided $M > 0$ D) $n_2 = 5$ provided $M > 5$

VIDEO SOLUTION



Instructions

Directions for the following three questions: Answer the following questions based on the statements given below:

- (i) There are three houses on each side of the road.
- (ii) These six houses are labeled as P, Q, R, S, T and U.
- (iii) The houses are of different colours, namely, Red, Blue, Green, Orange, Yellow and White.
- (iv) The houses are of different heights.
- (v) T, the tallest house, is exactly opposite to the Red coloured house.
- (vi) The shortest house is exactly opposite to the Green coloured house.
- (vii) U, the Orange coloured house, is located between P and S.
- (viii) R, the Yellow coloured house, is exactly opposite to P.
- (ix) Q, the Green coloured house, is exactly opposite to U.
- (x) P, the White coloured house, is taller than R, but shorter than S and Q.

Question 16

What is the colour of the tallest house?

- A Red
- B Blue
- C Green
- D Yellow
- E none of these

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Question 17

Which is the second tallest house?

- A P
- B S
- C Q
- D R
- E cannot be determined

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Question 18

What is the colour of the house diagonally opposite to the Yellow coloured house?

- A White
- B Blue
- C Green
- D Red
- E none of these

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Instructions

Directions for the following four questions: Answer the questions based on the following information.

A and B are two sets (e.g. A = Mothers, B = Women).

The elements that could belong to both the sets (e.g. women who are mothers) is given by the set $C = A \cap B$.

The elements which could belong to either A or B, or both, is indicated by the set $D = A \cup B$.

A set that does not contain any elements is known as a null set represented by ϕ (e.g. if none of the women in the set B is a mother, then $C = A \cap B$ is a null set, or $C = \phi$).

Let 'V' signify the set of all vertebrates, 'M' the set of all mammals, 'D' dogs, 'F' fish, 'A' alsatian and 'P', a dog named Pluto.

Question 19

If $P \cap A = \phi$ and $P \cup A = D$, then which of the following is true?

- A Pluto and alsatians are dogs
- B Pluto is an alsatian
- C Pluto is not an alsatian
- D Both A and C

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Question 20

If $Y = F \cap (D \cap V)$ is not a null set, it implies that

- A all fish are vertebrates
- B all dogs are vertebrates
- C some fish are dogs
- D None of these

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Question 21

Given that $X = M \cap D$ is such that $X = D$. Which of the following is true?

- A All dogs are mammals
- B Some dogs are mammals
- C $X = \phi$
- D All mammals are dogs

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CAT Percentile Predictor



Question 22

If $Z = (P \cap D) \cup M$, then

- A the elements of Z consist of Pluto, the dog, or any other mammal
- B Z implies any dog or mammal
- C Z implies Pluto or any dog that is a mammal
- D Z is a null set

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Instructions

Directions for the following two questions: Answer the following questions based on the information given below.

Elle is three times older than Yogesh. Zaheer is half the age of Wahida. Yogesh is older than Zaheer.

Question 23

Which of the following information will be sufficient to estimate Elle's age?

- A Zaheer is 10-year-old
- B Both Yogesh and Wahida are older than Zaheer by the same number of years
- C Both the above statements are needed
- D None of these

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Question 24

Which of the following can be inferred?

- A Yogesh is older than Wahida
- B Elle is older than Wahida
- C Elle may be younger than Wahida
- D None of these

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Instructions

DIRECTIONS for the following two questions: These questions are based on the situation given below: Seven university cricket players are to be honored at a special luncheon. The players will be seated on the dais along one side of a single rectangular table. A and G have to leave the luncheon early and must be seated at the extreme right end of the table, which is closest to the exit. B will receive the Man of the Match award and must be in the center chair. C and D who are bitter rivals for the position of wicket keeper, dislike one another and should be seated as far apart as possible. E and F are best friends and want to sit together.

Question 25

Which of the following pairs may not be seated together?

- A E & A
- B B & D
- C C & F
- D G & D

[VIEW SOLUTION](#)

Question 26

Which of the following may not be seated at either end of the table?

- A C
- B D
- C G
- D F

[VIEW SOLUTION](#)

Instructions

Answer the questions on the basis of the information given below.

These questions are based on the situation given below: A, B, C, D, E and F are a group of friends from a club. There are two housewives, one lecturer, one architect, one accountant and one lawyer in the group. There are two married couples in the group. The lawyer is married to D who is a housewife. No lady in the group is either

an architect or an accountant. C, the accountant, is married to F who is a lecturer. A is married to D and E is not a housewife.

Question 27

What is E?

- A Lawyer
- B Architect
- C Lecturer
- D Accountant

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Question 28

How many members of the group are male?

- A 2
- B 3
- C 4
- D None of these

[VIEW SOLUTION](#)

Instructions

There were seven elective courses - E1 to E7 - running in a specific term in a college. Each of the 300 students enrolled had chosen just one elective from among these seven. However, before the start of the term, E7 was withdrawn as the instructor concerned had left the college. The students who had opted for E7 were allowed to join any of the remaining electives. Also, the students who had chosen other electives were given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased; however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	1	4	2
	E2		34	8		2	2
	E3	2	6	25			2
	E4		3	2	14		4
	E5		5			30	
	E6		7	3		2	9
	E7	4	16	30	5	5	41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E4 was 3 more than that in E1.

- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

Question 29

How many elective courses among E1 to E6 had a decrease in their enrollments after the change process?

- A 4
- B 1
- C 2
- D 3

Each student was given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	1	4	2
	E2	8	8	8	2	2	2
	E3	2	6	25			2
	E4	3	2	14		4	
	E5	5				30	
	E6	7	3	2	2	9	
	E7	4	16	30	5	5	41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E1 was 3 more than that in E1.
- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

VIDEO SOLUTION

Question 30

After the change process, which of the following is the correct sequence of number of students in the six electives E 1 to E6?

- A 19, 76, 79, 21, 45, 60
- B 19, 76, 78, 22, 45, 60
- C 18, 76, 79, 23, 43, 61
- D 18, 76, 79, 21, 45, 61

Each student was given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	1	4	2
	E2	8	8	8	2	2	2
	E3	2	6	25			2
	E4	3	2	14		4	
	E5	5				30	
	E6	7	3	2	2	9	
	E7	4	16	30	5	5	41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E1 was 3 more than that in E1.
- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

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Question 31

Later, the college imposed a condition that if after the change of electives, the enrollment in any elective (other than E7) dropped to less than 20 students, all the students who had left that course will be required to re-enroll for that elective.

Which of the following is a correct sequence of electives in decreasing order of their final enrollments?

- A E2, E3, E6, E5, E1, E4
- B E3, E2, E6, E5, E4, E1
- C E2, E5, E3, E1, E4, E6
- D E2, E3, E5, E6, E1, E3

Each student is given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	3	4	2
	E2	8	8	8	2	2	2
	E3	2	6	25			
	E4	3	2	14			4
	E5	5				30	
	E6	7	3	2	2	9	
	E7	4	16	30	5	5	41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E1 was 3 more than that in E1.
- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

VIDEO SOLUTION

Question 32

After the change process, which course among E1 to E6 had the largest change in its enrollment as a percentage of its original enrollment?

- A E1
- B E2
- C E3
- D E6

Each student has one chance to change their choice. The table below tracks the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	3	4	2
	E2		34	8		2	
	E3	2	6	25			2
	E4		3	2	14		4
	E5					30	
	E6		7	3		2	9
	E7	4	14	30	5		41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E1 was 3 more than that in E1.
- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

VIDEO SOLUTION

Instructions

A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most two empty shelves can have consecutive numbers.

The following additional facts are known.

- A and B are to be placed in consecutively numbered shelves in increasing order.
- I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- K is to be placed in shelf number 16.
- L and J are items of the same type, while H is an item of a different type.
- C is a candy and is to be placed in a shelf preceded by two empty shelves.
- L is to be placed in a shelf preceded by exactly one empty shelf.

Question 33

In how many different ways can the items be arranged on the shelves?

- A 8
- B 4
- C 2
- D 1

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The following additional facts are known.

1. A and B are to be placed in consecutively numbered shelves in increasing order.
2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
4. K is to be placed in shelf number 16.
5. L and J are items of the same type, while H is an item of a different type.
6. C is a candy and is to be placed in a shelf preceded by two empty shelves.
7. L is to be placed in a shelf preceded by exactly one empty shelf.

Handwritten notes:

- 12 items → A, B, ..., L
- 16 shelves
- 5 Biscuits
- 3 Candies
- 4 Savouries
- I, J



CAT Daily Targets

Step by step, heading towards 99+ percentile



Question 34

Which of the following statements is necessarily true?

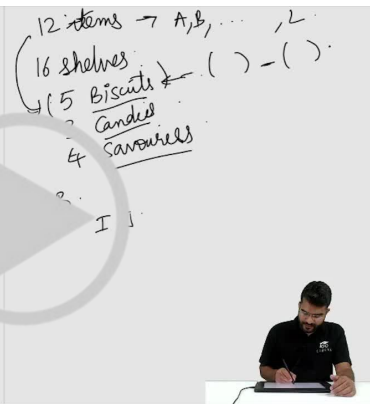
- A All biscuits are kept before candies.
- B There are two empty shelves between the biscuits and the candies.
- C All candies are kept before biscuits.
- D There are at least four shelves between items B and C.

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A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most two empty shelves can have consecutive numbers.

The following additional facts are known.

- 1. A and B are to be placed in consecutively numbered shelves in increasing order.
- 2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- 3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- 4. K is to be placed in shelf number 16.
- 5. L and J are items of the same type, while H is an item of a different type.
- 6. C is a candy and is to be placed in a shelf preceded by two empty shelves.



Handwritten notes on the right side of the video player:

- 12 items → A, B, ..., L
- 16 shelves → () - ()
- 5 Biscuits
- 3 Candies
- 4 Savouries

Question 35

Which of the following items is not a type of biscuit?

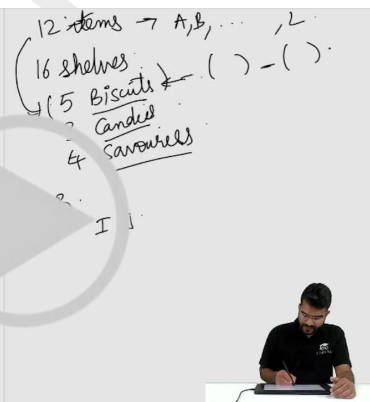
- A L
- B A
- C B
- D G

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A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most two empty shelves can have consecutive numbers.

The following additional facts are known.

- 1. A and B are to be placed in consecutively numbered shelves in increasing order.
- 2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- 3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- 4. K is to be placed in shelf number 16.
- 5. L and J are items of the same type, while H is an item of a different type.
- 6. C is a candy and is to be placed in a shelf preceded by two empty shelves.



Handwritten notes on the right side of the video player:

- 12 items → A, B, ..., L
- 16 shelves → () - ()
- 5 Biscuits
- 3 Candies
- 4 Savouries

Question 36

Which of the following can represent the numbers of the empty shelves in a possible arrangement?

- A 1, 7, 11, 12
- B 1, 5, 6, 12
- C 1, 2, 6, 12
- D 1, 2, 8, 12

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The following additional facts are known:

1. A and B are to be placed in consecutively numbered shelves in increasing order.
2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
4. K is to be placed in shelf number 16.
5. L and J are items of the same type, while H is an item of a different type.
6. C is a candy and is to be placed in a shelf preceded by two empty shelves.

Handwritten notes: 12 items → A, B, ..., L. 16 shelves. (5 Biscuits) - () - (). 3 Candies. 4 Savouries. I J.

FREE

CAT Sectional Tests

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Take Here

Instructions

Directions for the next 2 questions: There are three bottles of water, A, B, C, whose capacities are 5 litres, 3 litres, and 2 litres respectively. For transferring water from one bottle to another and to drain out the bottles, there exists a piping system. The flow through these pipes is computer controlled. The computer that controls the flow through these pipes can be fed with three types of instructions, as explained below:

Instruction type	Explanation for the instruction
FILL (X, Y)	Fill bottle labeled X from the water in bottle labeled Y, where the remaining capacity of X is less than or equal to the amount of water in Y.
EMPTY (X, Y)	Empty out water in bottle labeled X into bottle labeled Y, where the amount of water in X is less than or equal to remaining capacity of Y.
DRAIN (X)	Drain out all the water contained in bottle labeled X.

Initially, A is full with water, and B and C are empty.

Question 37

After executing a sequence of three instructions, bottle A contains one litre of water. The first and the third of these instructions are shown below:

First instruction: FILL (C, A)

Third instruction: FILL (C, A)

Then which of the following statements about the instruction is true?

- A The second instruction is FILL (B, A)
- B The second instruction is EMPTY (C, B)
- C The second instruction transfers water from B to C
- D The second instruction involves using the water in bottle A

[VIEW SOLUTION](#)

Question 38

Consider the same sequence of three instructions 'and the same initial state mentioned above. Three more instructions are added at the end of the above sequence to have A contain 4 litres of water. In this total sequence of six instructions, the fourth one is DRAIN (A). This is the only DRAIN instruction in the entire sequence. At the end of the execution of the above sequence, how much water (in litres) is contained in C?

- A One
- B Two
- C Zero
- D None of these

[VIEW SOLUTION](#)

Instructions

Seven children, Aarav, Bina, Chirag, Diya, Eshan, Farhan, and Gaurav, are sitting in a circle facing inside (not necessarily in the same order) and playing a game of 'Passing the Buck'.

The game is played over 10 rounds. In each round, the child holding the Buck must pass it directly to a child sitting in one of the following positions:

- Immediately to the left;
- Immediate to the right;
- Second to the left; or
- Second to the right.

The game starts with Bina passing the Buck and ends with Chirag receiving the Buck. The table below provides some information about the pass types and the child receiving the Buck. Some information is missing and labelled as '?'.
?

Round	Pass Type	Received by
1	Immediately to the left	Aarav
2	Second to the right	?
3	Immediately to the right	Diya
4	?	?
5	?	Aarav
6	Second to the left	?
7	Immediately to the left	Gaurav
8	Immediately to the left	?
9	?	Farhan
10	?	Chirag

Question 39

For which of the following children is it possible to determine how many times they received the Buck?

- A Farhan
- B Eshan
- C Bina
- D Gaurav

[VIEW SOLUTION](#)



Question 40

Who is sitting third to the left of Eshan?

- A Gaurav
- B Divya
- C Chirag
- D Aarav

[VIEW SOLUTION](#)

Question 41

For which of the following pass types can the total number of occurrences be uniquely determined?

- A Immediately to the left
- B Second to the right
- C Immediately to the right
- D Second to the left

[VIEW SOLUTION](#)

Question 42

Who is sitting immediately to the right of Bina?

- A Aarav
- B Eshan
- C Farhan
- D Chirag

[VIEW SOLUTION](#)

Instructions

For the following questions answer them individually

Question 43

Five boys went to a store to buy sweets. One boy had Rs. 40. Another boy had Rs. 30. Two other boys had Rs. 20 each. The remaining boy had Rs. 10. Below are some more facts about the initial and final cash positions.

- I. Alam started with more than Jugraj.
- II. Sandeep spent Rs. 1.50 more than Daljeet.
- III. Ganesh started with more money than just only one other person.
- IV. Daljeet started with $\frac{2}{3}$ of what Sandeep started with.
- V. Alam spent the most, but did not end with the least.
- VI. Jugraj spent the least and ended with more than Alam or Daljeet.
- VII. Ganesh spent Rs. 3.50.
- VIII. Alam spent 10 times than what Ganesh did.

In the choices given below, all statements except one are false. Which one of the following statements can be true?

- A Alam started with Rs. 40 and ended with Rs. 9.50
- B Sandeep started with Rs. 30 and ended with Re. 1
- C Ganesh started with Rs. 20 and ended with Rs. 4
- D Jugraj started with Rs. 10 and ended with Rs. 7

[VIEW SOLUTION](#)

Question 44

At a village mela, the following six nautankis (plays) are scheduled as shown in the table below.

No.	Nautanki	Duration	Show Times
1	Sati Savitri	1 hr	9 am and 2 pm
2	Joru ka Ghulam	1 hr	10:30 am and 11:30 am
3	Sundar Kand	30 min	10 am and 11 am
4	Veer Abhimanyu	1 hr	10 am and 11 am
5	Reshma aur Shera	1 hr	9:30 am, 12 noon and 2 pm
6	Jhansi ki Rani	30 min	11 am and 1:30 pm

You wish to see all the six nautankis. Further, you wish to ensure that you get a lunch break from 12.30 p.m. to 1.30 p.m. Which of the following ways can you do this?

- A Sati Savitri is viewed first; Sundar Kand is viewed third, and Jhansi ki Rani is viewed last
- B Sati Savitri is viewed last; Veer Abhimanyu is viewed third, and Reshma aur Shera is viewed first
- C Sati Savitri is viewed first; Sundar Kand is viewed third, and Joru ka Ghulam is viewed fourth
- D Veer Abhimanyu is viewed third; Reshma aur Shera is viewed fourth, and Jahansi ki Rani is viewed fifth

[VIEW SOLUTION](#)

Question 45

Three children won the prizes in the Bournvita Quiz contest. They are from the schools: Loyola, Convent and Little Flowers, which are located at different cities. Below are some of the facts about the schools, the children and the city they are from.

- I. One of the children is Bipin.
- II. Loyola School's contestant did not come first.
- III. Little Flower's contestant was named Riaz.
- IV. Convent School is not in Hyderabad.
- V. The contestant from Pune is not from Loyola School.
- VI. The contestant from Bangalore did not come first.
- VII. Convent School's contestant's name is not Balbir.
- VIII. The contestant from Pune came in third.

Which of the following statements is true?

- A 1st prize: Riaz (Little Flowers), 2nd prize: Bipin (Convent), 3rd prize: Balbir (Loyola)
- B 1st prize: Bipin (Convent), 2nd prize: Riaz (Little Flowers), 3rd prize: Balbir (Loyola)
- C 1st prize: Riaz (Little Flowers), 2nd prize: Balbir (Loyola), 3rd prize: Bipin (Convent)
- D 1st prize: Bipin (Convent), 2nd prize: Balbir (Loyola), 3rd prize: Riaz (Little Flowers)

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100+ CAT Quant Questions



Question 46

The Bannerjees, the Sharmas, and the Pattabhiramans each have a tradition of eating Sunday lunch as a family. Each family serves a special meal at a certain time of day. Each family has a particular set of chinaware used for this meal. Use the clues below to answer the following question.

- . The Sharma family eats at noon.
- . The family that serves fried brinjal uses blue chinaware.
- . The Bannerjee family eats at 2 o'clock.
- . The family that serves sambar does not use red chinaware.
- . The family that eats at 1 o'clock serves fried brinjal.
- . The Pattabhiraman family does not use white chinaware.
- . The family that eats last likes makkai-ki-roti.

Which one of the following statements is true?

- A The Bannerjees eat makkai-ki-roti at 2 o'clock, the Sharmas eat fried brinjal at 12 o'clock and the Pattabhiramans eat sambar from red chinaware
- B The Sharmas eat sambar served in white chinaware, the Pattabhiramans eat fried brinjal at 1 o'clock, and the Bannerjees eat makkai-ki-roti served in blue chinaware
- C The Sharmas eat sambar at noon, the Pattabhiramans eat fried brinjal served in blue chinaware, and the Bannerjees eat makkai-ki-roti served in red chinaware

- D** The Bannerjees eat makkai-ki-roti served in white chinaware, the Sharmas eat fried brinjal at 12 o'clock and the Pattabhiramans eat sambar from red chinaware

 [VIEW SOLUTION](#)

Question 47

Albert, David, Jerome and Tommy were plucking mangoes in a grove to earn some pocket money during the summer holidays. Their earnings were directly related to the number of mangoes plucked and had the following relationship: Jerome got less money than Tommy. Jerome and Tommy together got the same amount as Albert and David taken together. Albert and Tommy together got less than David and Jerome taken together. Who earned the most pocket money? Who plucked the least number of mangoes?

- A** David, Jerome
- B** David, Albert
- C** Jerome, Tommy
- D** Jerome, Albert

 [VIEW SOLUTION](#)

Question 48

Eight people carrying food baskets are going for a picnic on motorcycles.

Their names are A, B, C, D, E, F, G, and H. They have 4 motorcycles M1, M2, M3 and M4 among them. They also have 4 food baskets O, P, Q and R of different sizes and shapes and each can be carried only on motorcycles M1, M2, M3 and M4 respectively.

No more than 2 persons can travel on a motorcycle and no more than one basket can be carried on a motorcycle. There are 2 husband-wife pairs in this group of 8 people and each pair will ride on a motorcycle together.

C cannot travel with A or B. E cannot travel with B or F. G cannot travel with F, or H, or D.

The husband-wife pairs must carry baskets O and P. Q is with A and P is with D.

F travels on M1 and E travels on M2 motorcycles.

G is with Q, and B cannot go with R.

Who is travelling with H?

- A** A
- B** B
- C** C
- D** D

cracku

Eight people carrying food baskets are going for a picnic on motorcycles. Their names are A, B, C, D, E, F, G, and H. They have 4 motorcycles M1, M2, M3 and M4 among them. They also have 4 food baskets O, P, Q and R of different sizes and shapes and each can be carried only on motorcycles M1, M2, M3 and M4 respectively.

No more than 2 persons can travel on a motorcycle and no more than one basket can be carried on a motorcycle. There are 2 husband-wife pairs in this group of 8 people and each pair will ride on a motorcycle together.

C cannot travel with A or B. E cannot travel with B or F. G cannot travel with F, C or D.

The husband-wife pairs must carry baskets O and P. H is with A and P is with D.

F travels on M1 and E travels on M2 motorcycle.

G is with Q, and B cannot go with R.

Who is travelling with H?

A) A
B) B
C) C
D) D



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100+ CAT DILR Questions



Question 49

Mrs Ranga has three children and has difficulty remembering their ages and months of their birth. The clue below may help her remember.

- . The boy, who was born in June, is 7 years old.
- . One of the children is 4 years old but it was not Anshuman.
- . Vaibhav is older than Suprita.
- . One of the children was born in September, but it was not Vaibhav.
- . Suprita's birthday is in April.
- . The youngest child is only 2-year-old.

Based on the above clues, which one of the following statements is true?

- A** Vaibhav is the oldest, followed by Anshuman who was born in September, and the youngest is Suprita who was born in April
- B** Anshuman is the oldest being born in June, followed by Suprita who is 4-year-old, and the youngest is Vaibhav who is 2-year-old
- C** Vaibhav is the oldest being 7-year-old, followed by Suprita who was born in April, and the youngest is Anshuman who was born in September
- D** Suprita is the oldest who was born in April, followed by Vaibhav who was born in June, and Anshuman who was born in September

VIEW SOLUTION

Question 50

Six persons are playing a card game sitting around a circular table. Suresh is facing Raghubir who is to the left of Ajay and to the right of Pramod. Ajay is to the left of Dhiraj. Yogendra is to the left of Pramod. If Dhiraj exchanges his seat with Yogendra and Pramod exchanges with Raghubir, who will be sitting to the left of Dhiraj?

- A** Yogendra
- B** Raghubir

C Suresh

D Ajay

[VIEW SOLUTION](#)

Question 51

Four students — Ashish, Dhanraj, Felix and Sameer sat for the Common Entrance Exam for Management (CEEM). Each of the students has one profession among the four: Doctor, Chartered Accountant, Engineer, and Economist. One student got admission offers from three NIMs (National Institutes of Management), another from two NIMs, the third from one NIM, while the fourth got none. Below are some of the facts about who got admission offers from how many NIMs and what is their educational background.

- I. The one who is an engineer didn't get as many admissions as Ashish.
- II. The one who got the offer for admissions in two NIMs isn't Dhanraj nor is he a chartered accountant.
- III. Sameer is an economist.
- IV. Dhanraj isn't an engineer and received more admission offers than Ashish.
- V. The doctor got the most number of admission offers.

Which one of the following statements is necessarily true?

- A Ashish is a chartered accountant and got offer for admission in three NIMs.
- B Dhanraj is a doctor and got admission offer in one NIM.
- C Sameer is an economist who got admission offers in two NIMs.
- D Felix who is not an engineer did not get any offer for admission.

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100+ CAT VARC Questions



Question 52

Abraham, Border, Charlie, Dennis and Elmer and their respective wives recently dined together and were seated at a circular table. The seats were so arranged that men and women alternated and each woman was three places distant from her husband. Mrs. Charlie sat to the immediate left of Mr. Abraham. Mrs. Elmer sat two places to the right of Mrs. Border. Who sat to the right of Mr. Abraham?

- A Mrs. Dennis
- B Mrs. Elmer
- C Mrs. Border
- D Mrs. Border or Mrs. Dennis

cracku

Abraham, Border, Charlie, Dennis and Elmer and their respective wives recently dined together and were seated at a circular table. The seats were so arranged that men and women alternated and each woman was three places distant from her husband. Mrs. Charlie sat to the left of Mr. Abraham. Mrs. Elmer sat two places to the right of Mrs. Border. Who sat to the right of Mr. Abraham?

A) Mrs. Dennis
C) Mrs. Border

B) Mrs. Elmer
Mrs. Border or Mrs. Dennis

VIDEO SOLUTION

Question 53

I happened to be the judge in the all India Essay Competition on Nylon Dying, organized some time back by a dyestuff firm. Mill technicians were eligible to enter the competition. My work was simplified in assessing the essays, which had to be done under five heads-Language, Coherence, Subject Matter, Machinery and Recent Developments. Integral marks were to be given out of a maximum of 20 under each head. There were only five entries. The winner got 90 marks. Only a single person can be made the winner. Akhila got 13 in Coherence and Divya 10 in Machinery. Bhanu.s total was less than Akhila.s. Charulata has sent an entry. Ela had got as many marks as Divya. None got 20 under any head. Who was the winner?

- A Divya
- B Charulata
- C Ela
- D Bhanu

VIEW SOLUTION

Question 54

In a 'keep-fit' gymnasium class there are 15 females enrolled in a weight-loss programme. They all have been grouped in any one of the five weight-groups W1, W2, W3, W4, or W5. One instructor is assigned to one weight-group only.

Sonali, Shalini, Shubhra and Shahira belong to the same weight-group.

Sonali and Rupa are in one weight-group, Rupali and Renuka are also in one weight-group.

Rupa, Radha, Renuka, Ruchika, and Ritu belong to different weight groups.

Somya cannot be with Ritu, and Tara cannot be with Radha.

Komal cannot be with Radha, Somya, or Ritu.

Shahira is in W1 and Somya is in W4 with Ruchika.

Sweta and Jyotika cannot be with Rupali, but are in a weight-group with total membership of four.

No weight-group can have more than five or less than one member.

Amita, Babita, Chandrika, Deepika and Elina are instructors of weight-groups with membership sizes 5, 4, 3, 2 and 1 respectively. Who is the instructor of Radha?

- A Babita
- B Elina
- C Chandrika

CAT Expected Questions PDF

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Question 55

There is a vertical stack of books marked 1, 2 and 3 on Table-A, with 1 at the bottom and 3 on top. These are to be placed vertically on Table-B with 1 at the bottom and 2 on the top, by making a series of moves from one table to the other. During a move, the topmost book, or the topmost two books, or all the three, can be moved from one of the tables to the other. If there are any books on the other table, the stack being transferred should be placed, on top of the existing books, without changing the order of books in the stack that is being moved in that move. If there are no books on the other table, the stack is simply placed on the other table without disturbing the order of books in it. What is the minimum number of moves in which the above task can be accomplished?

- A One
- B Two
- C Three
- D Four

Question 56

Five persons with names P, M, U, T and X live separately in any one of the following: a palace, a hut, a fort, a house or a hotel. Each one likes two different colours from among the following: blue, black, red, yellow and green. U likes red and blue. T likes black. The person living in a palace does not like black or blue. P likes blue and red. M likes yellow. X lives in a hotel. M lives in a:

- A hut
- B palace
- C fort
- D house

Question 57

Persons X, Y, Z and Q live in red, green, yellow or blue coloured houses placed in a sequence on a street. Z lives in a yellow house. The green house is adjacent to the blue house. X does not live adjacent to Z. The yellow house is the only house in between the green and red houses. The colour of the house X lives in is:

- A blue
- B green
- C red

D not possible to determine

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CAT Important Topics



Instructions

A, B, C and D are to be seated in a row. But C and D cannot be together. Also B cannot be at the third place.

Question 58

If A and B are together, then which of the following must be necessarily true?

- A C is not at the first place
- B A is at the third place
- C D is at the first place
- D C is at the first place

[VIEW SOLUTION](#)

Question 59

If A is not at the third place, then which of the following options does C have?

- A The first place only
- B The third place only
- C The first and third place only
- D Any of the places

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Question 60

Which of the following must be false?

- A A is at the first place
- B A is at the second place
- C A is at the third place
- D A is at the fourth place

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CAT WhatsApp Group



Instructions

Direction for the following four questions: Answer the questions based on the following information. Mr Bankatlal acted as a judge for the beauty contest. There were four participants, viz. Ms Andhra Pradesh, Ms Uttar Pradesh, Ms West Bengal and Ms Maharashtra. Mrs Bankatlal, who was very anxious about the result, asked him about it as soon as he was back home. Mr Bankatlal just told that the one who was wearing the yellow saree won the contest. When Mrs Bankatlal pressed for further details, he elaborated as follows: All of them were sitting in a row. All of them wore sarees of different colours, viz. green, yellow, white, red. There was only one runner-up and she was sitting beside Ms. Maharashtra. The runner-up was wearing the green saree. Ms West Bengal was not sitting at the ends and was not the runner up. The winner and the runner-up are not sitting adjacent to each other. Ms Maharashtra was wearing white saree. Ms Andhra Pradesh was not wearing the green saree. Participants wearing yellow saree and white saree were at the ends.

Question 61

Who was the runner-up?

- A Ms Andhra Pradesh
- B Ms West Bengal
- C Ms Uttar Pradesh
- D Ms Maharashtra

 [VIEW SOLUTION](#)

Question 62

Who wore the red saree?

- A Ms Andhra Pradesh
- B Ms West Bengal
- C Ms Uttar Pradesh
- D Ms Maharashtra

 [VIEW SOLUTION](#)

Question 63

Which saree was worn by Ms Andhra Pradesh?

- A Yellow
- B Red
- C Green
- D White

 [VIEW SOLUTION](#)



CAT Mock Test

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Question 64

Ms. West Bengal was sitting adjacent to

- A Ms Andhra Pradesh and Ms Maharashtra
- B Ms Uttar Pradesh and Ms Maharashtra
- C Ms Andhra Pradesh and Ms Uttar Pradesh
- D Ms Uttar Pradesh

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Instructions

A round table has seven chairs around it. The chairs are numbered 1 through 7 in a clockwise direction. Four friends, Aslam, Bashir, Chhavi, and Davies, sit on four of the chairs. In the starting position, Aslam and Chhavi are sitting next to each other, while for Bashir as well as Davies, there are empty chairs on either side of the chairs that are sitting on.

The friends take turns moving either clockwise or counterclockwise from their chair. The friend who has to move in a turn occupies the first empty chair in whichever direction (s)he chooses to move. Aslam moves first (Turn 1), followed by Bashir, Chhavi, and Davies (Turns 2, 3, and 4, respectively). Then Aslam moves again followed by Bashir, and Chhavi (Turns 5, 6, and 7, respectively).

The following information is known

1. The four friends occupy adjacent chairs only at the end of Turn 2 and Turn 6.
2. Davies occupies Chair 2 after Turn 1 and Chair 4 after Turn 5, and Chhavi occupies Chair 7 after Turn 2.

Question 65

Which of the chairs are occupied at the end of Turn 6?

- A Chairs numbered 4, 5, 6, and 7
- B Chairs numbered 1, 2, 3, and 4
- C Chairs numbered 2, 3, 4, and 5
- D Chairs numbered 1, 2, 6, and 7

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Question 66

Who sits on the chair numbered 4 at the end of Turn 3?

- A Bashir
- B Chhavi
- C Davies
- D No one

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Question 67

Which of the following BEST describes the friends sitting on chairs adjacent to the one occupied by Bashir at the end of Turn 7?

- A Chhavi only
- B Davies only
- C Chhavi and Davies
- D Aslam and Chhavi

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Question 68

What is the number of the chair initially occupied by Bashir?

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Instructions

Princess, Queen, Rani and Samragini were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancers. The dancers performed their second items in the same sequence of their performance of their first items.

The following additional facts are known.

- i) No composer who assigned item to Princess, assigned any item to Queen.
- ii) No composer who assigned item to Rani, assigned any item to Samragini.
- iii) The first performance was by Princess; this item was assigned by Badal.
- iv) The last performance was by Rani; this item was assigned by Gagan.
- v) The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Question 69

The sixth performance was composed by:

- A Badal
- B Dyu
- C Ashman
- D Gagan

Princess, Queen, Rani and Samragni were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragni.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Finalist		Music Composers	
1	.		
2	.		
3	.		
4	.		
5	.		
6	.		
7	.		
8	.		



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Question 70

Which pair of performances were composed by the same composer?

- The first and the seventh
- The third and the seventh
- The second and the sixth
- The first and the sixth

Princess, Queen, Rani and Samragni were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragni.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Finalist		Music Composers	
1	.		
2	.		
3	.		
4	.		
5	.		
6	.		
7	.		
8	.		



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Question 71

Which of the following is true?

- The second performance was composed by Dyu.
- The third performance was composed by Dyu.
- The third performance was composed by Ashman.
- The second performance was composed by Gagan.

cracku

Princess, Queen, Rani and Samragini were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragini.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Handwritten notes on a whiteboard:
 P, A, R, S. A, B, G, D.
 Finalist Music Composers.
 A table with 8 rows and 2 columns. The first column contains numbers 1 through 8. The second column contains dots. The first four rows are grouped under 'Finalist' and the last four rows under 'Music Composers'.



VIDEO SOLUTION

Question 72

Which of the following is FALSE?

- A Samragini did not perform in any item composed by Ashman.
- B Princess did not perform in any item composed by Dyu.
- C Rani did not perform in any item composed by Badal.
- D Queen did not perform in any item composed by Gagan.

cracku

Princess, Queen, Rani and Samragini were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragini.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Handwritten notes on a whiteboard:
 P, A, R, S. A, B, G, D.
 Finalist Music Composers.
 A table with 8 rows and 2 columns. The first column contains numbers 1 through 8. The second column contains dots. The first four rows are grouped under 'Finalist' and the last four rows under 'Music Composers'.



VIDEO SOLUTION

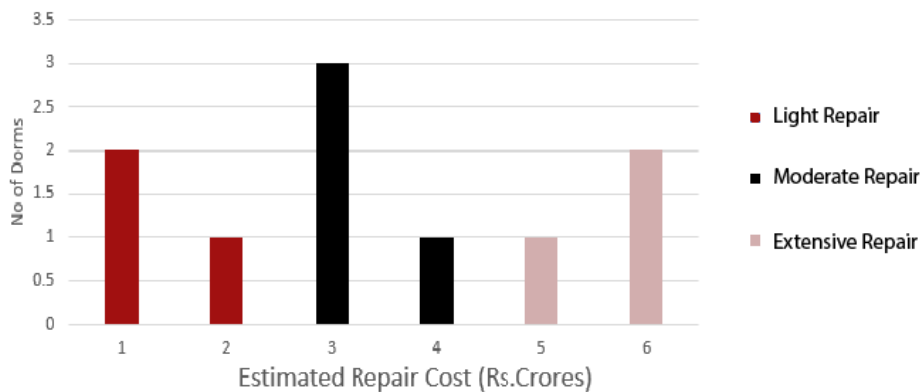


CAT Syllabus PDF



Instructions

At a management school, the oldest M dorms, numbered 1 to 10, need to be repaired urgently. This following diagram represents the estimated repair costs (in Rs. Crores for, the 10 dorms. For any dorm, the estimated repair cost (in Rs. Crores) is an integer. Repairs with estimated cost Rs. 1 or 2 Crores are considered light repairs, repairs with estimated cost Rs. 3 or 4 are considered moderate repairs and repairs with estimated cost Rs. 5 or 6 Crores are considered extensive repairs.



Further, the following information is known.

1. Odd-numbered dorms do not need light repair; even-numbered dorms do not need moderate repair and dorms, whose numbers are divisible by 3, do not need extensive repair.
2. Dorms 4 to 9 all need different repair costs, with Dorm 7 needing the maximum and Dorm 8 needing the minimum.

Question 73

Which of the following is NOT necessarily true?

- A Dorm 1 needs a moderate repair
- B Dorm 5 repair will cost no more than Rs. 4 Crores
- C Dorm 7 needs an extensive repair
- D Dorm 10 repair will cost no more than Rs. 4 Crores

At a management school, the oldest 10 dorms, numbered 1 to 10, need to be repaired urgently. This following diagram represents the estimated repair costs (in Rs. Crores) for the 10 dorms. For any dorm, the estimated repair cost (in Rs. Crores) is an integer. Repairs with estimated cost Rs. 1 or 2 Crores are considered light repairs, repairs with estimated cost Rs. 3 or 4 Crores are considered moderate repairs, repairs with estimated cost Rs. 5 or 6 Crores are considered extensive repairs.

Further, the following information is known.

1. Odd-numbered dorms do not need light repair; even-numbered dorms do not need moderate repair and dorms, whose numbers are divisible by 3, do not need extensive repair.
2. Dorms 4 to 9 all need different repair costs, with Dorm 7 needing the maximum and Dorm 8 needing the minimum.

VIDEO SOLUTION

Question 74

What is the total cost of repairing the odd-numbered dorms (in Rs. Crores)?



VIDEO SOLUTION

Question 75

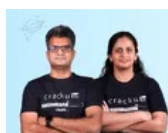
Suppose further that:

1. 4 of the 10 dorms needing repair are women's dorms and need a total of Rs. 20 Crores for repair.
2. Only one of Dorms 1 to 5 is a women's dorm.

What is the cost for repairing Dorm 9 (in Rs. Crores)?



VIDEO SOLUTION



CAT Formula PDF



Question 76

Suppose further that:

1. 4 of the 10 dorms needing repair are women's dorms and need a total of Rs. 20 Crores for repair.
2. Only one of Dorms 1 to 5 is a women's dorm.

Which of the following is a women's dorm?

- A Dorm 2
- B Dorm 5
- C Dorm 8
- D Dorm 10

At a management school, the oldest M dorms, numbered 1 to 50, need to be repaired urgently. This following diagram represents the estimated repair costs (in Rs. Crores) for the 50 dorms. For any dorm, the estimated repair cost (in Rs. Crores) is an integer. Repairs with estimated cost Rs. 1 or 2 Crores are considered light repairs, repairs with estimated cost Rs. 3 or 4 Crores are considered moderate repairs and repairs with estimated cost Rs. 5 or 6 Crores are considered extensive repairs.

Further, the following information is known:

1. Odd-numbered dorms do not need light repair; even-numbered dorms do not need moderate repair and dorms, whose numbers are divisible by 3, do not need extensive repair.
2. Dorms 4 to 9 all need different repair costs, with Dorm 7 needing the maximum and Dorm 8 needing the minimum.

Handwritten notes on the right side of the screen show calculations for repair costs:

light: $2 - 1 = 1$
moderate: $3 - 3 = 0$
extensive: $5 - 1 = 4$
 $6 - 2 = 4$
 $\frac{10}{10}$

VIDEO SOLUTION

Instructions

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal: These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- Tea from Ooty got the highest rating, but it was not in Cup 3.
- The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- Only two cups got ratings in even numbers.
- Cup 2 got the minimum rating and this rating was an even number.
- Tea in Cup 3 got a higher rating than that in Cup 1.
- The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Question 77

What was the second highest rating given?

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal: These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- Tea from Ooty got the highest rating, but it was not in Cup 3.
- The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- Only two cups got ratings in even numbers.
- Cup 2 got the minimum rating and this rating was an even number.
- Tea in Cup 3 got a higher rating than that in Cup 1.
- The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-10)
1		
2		
3		
4		
5		
6	Himachal	

VIDEO SOLUTION

Question 78

What was the number of the cup that contained tea from Ooty?

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 50. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- 2. Tea from Ooty got the highest rating, but it was not in Cup 3.
- 3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- 4. Only two cups got ratings in even numbers.
- 5. Cup 2 got the minimum rating and this rating was an even number.
- 6. Tea in Cup 3 got a higher rating than that in Cup 1.
- 7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-50)
1		
2		
3		
4		
5		
6	Himachal	

VIDEO SOLUTION

IIM Call Predictor
Accurate analysis of your profile



Question 79

If the tea from Munnar did not get the minimum rating, what was the rating of the tea from Wayanad?

- A 3
- B 5
- C 1
- D 6

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 50. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- 2. Tea from Ooty got the highest rating, but it was not in Cup 3.
- 3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- 4. Only two cups got ratings in even numbers.
- 5. Cup 2 got the minimum rating and this rating was an even number.
- 6. Tea in Cup 3 got a higher rating than that in Cup 1.
- 7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-50)
1		
2		
3		
4		
5		
6	Himachal	

VIDEO SOLUTION

Question 80

If cups containing teas from Wayanad and Ooty had consecutive numbers, which of the following statements may be true?

- A Cup 5 contains tea from Assam
- B Cup 1 contains tea from Darjeeling
- C Tea from Wayanad has got a rating of 6

D Tea from Darjeeling got the minimum rating

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

- 1. Cup 6 contained tea from Himachal.
- 2. Tea from Ooty got the highest rating, but it was not in Cup 3.
- 3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- 4. Only two cups got ratings in even numbers.
- 5. Cup 2 got the minimum rating and this rating was an even number.
- 6. Tea in Cup 3 got a higher rating than that in Cup 1.
- 7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-10)
1		
2		
3		
4		
5		
6	Himachal	



VIDEO SOLUTION

Instructions

Eight friends: Ajit, Byomkesh, Gargi, Jayanta, Kikira, Manik, Prodosh and Tapesha are going to Delhi from Kolkata by a flight operated by Cheap Air. In the flight, sitting is arranged in 30 rows, numbered 1 to 30, each consisting of 6 seats, marked by letters A to F from left to right, respectively. Seats A to C are to the left of the aisle (the passage running from the front of the aircraft to the back), and seats D to F are to the right of the aisle. Seats A and F are by the windows and referred to as Window seats, C and D are by the aisle and are referred to as Aisle seats while B and E are referred to as Middle seats. Seats marked by consecutive letters are called consecutive seats (or seats next to each other). A seat number is a combination of the row number, followed by the letter indicating the position in the row; e.g., 1A is the left window seat in the first row, while 12E is the right middle seat in the 12th row.

Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as those have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

1. The eight friends were seated in six different rows.
2. They occupied 3 Window seats, 4 Aisle seats and 1 Middle seat.
3. Seven of them had to pay extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
4. Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
5. Gargi was sitting next to Kikira, and Manik was sitting next to Jayanta.
6. Prodosh and Tapesha were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but they paid different amounts for their choices of seat. One of these amounts may be zero.

Question 81

In which row was Manik sitting?

- A 10
- B 11
- C 12
- D 13

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.
 Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seventeen paid 3 Window seats, 4 Aisle seats and 1 Middle seat.
- Seven of them had to pay extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Ajit and Gargi were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kira, and Manik was sitting next to Jayanta.
- Sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different seat. One of these amounts may be zero.

→ 1 - Rs. 1000
 2 - 300 for window
 500 for aisle

VIDEO SOLUTION

CAT Percentile Predictor



Question 82

How much extra did Gargi pay for her choice of seat?

- A 0
- B Rs. 300
- C Rs. 400
- D Rs. 1000

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.
 Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seventeen paid 3 Window seats, 4 Aisle seats and 1 Middle seat.
- Seven of them had to pay extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Ajit and Gargi were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kira, and Manik was sitting next to Jayanta.
- Sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different seat. One of these amounts may be zero.

→ 1 - Rs. 1000
 2 - 300 for window
 500 for aisle

VIDEO SOLUTION

Question 83

How much extra did Jayanta pay for his choice of seat?

- A Rs. 300
- B Rs. 400
- C Rs. 500
- D Rs. 1000

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.
 Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seven of them paid extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Alit and Tapesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kikira, and Manik was sitting next to Jayanta.
- Three of them were sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different amounts for their choices of seat. One of these amounts may be zero.

→ 1 - 300 for window
 → 2 - 500 for aisle

VIDEO SOLUTION

Question 84

Who among the following did not pay any extra amount for his his/her choice of seat?

- A Kikira
- B Manik
- C Gargi
- D Tapesh

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.
 Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seven of them paid extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Alit and Tapesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kikira, and Manik was sitting next to Jayanta.
- Three of them were sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different amounts for their choices of seat. One of these amounts may be zero.

→ 1 - 300 for window
 → 2 - 500 for aisle

VIDEO SOLUTION

CAT Score Calculator

Instructions

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

- The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
- The number of educationalists in the teaching committee is less than the number of educationalists in the

research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.

3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

Question 85

Based on the given information, which of the following statements MUST be FALSE?

- A In the teaching committee the number of educationalists is equal to the number of politicians
- B In the administration committee the number of bureaucrats is equal to the number of educationalists
- C The size of the research committee is less than the size of the teaching committee
- D The size of the research committee is less than the size of the administration committee

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
P				
Total				24

[VIDEO SOLUTION](#)

Question 86

What is the number of bureaucrats in the administration committee?

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
P				
Total				24

[VIDEO SOLUTION](#)

Question 87

What is the number of educationalists in the research committee?

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
P				
Total				24

VIDEO SOLUTION

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Question 88

Which of the following CANNOT be determined uniquely based on the given information?

- A The size of the teaching committee
- B The size of the research committee
- C The total number of bureaucrats in the three committees
- D The total number of educationalists in the three committees

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

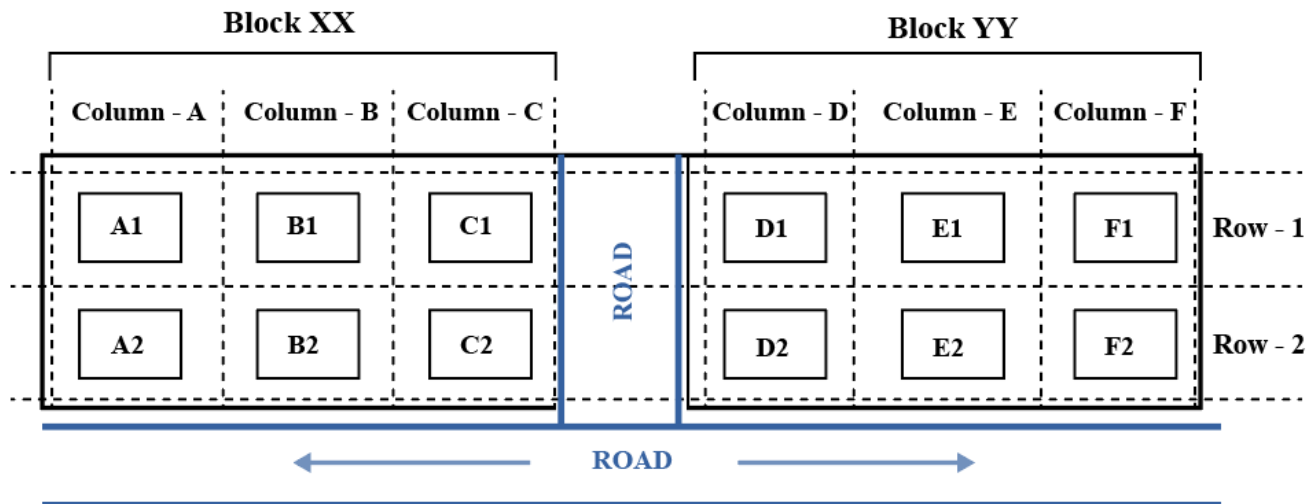
1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
P				
Total				24

VIDEO SOLUTION

Instructions

The schematic diagram below shows 12 rectangular houses in a housing complex. House numbers are mentioned in the rectangles representing the houses. The houses are located in six columns - Column-A through Column-F, and two rows - Row-1 and Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.



Some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The road adjacency value of a house is the number of its sides adjacent to a road. For example, the road adjacency values of C2, F2, and B1 are 2, 1, and 0, respectively. The neighbour count of a house is the number of sides of that house adjacent to occupied houses in the same block. For example, E1 and C1 can have the maximum possible neighbour counts of 3 and 2, respectively.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is also known.

1. The maximum quoted price of a house in Block XX is Rs. 24 lakhs. The minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column-E.
2. Row-1 has two occupied houses, one in each block.
3. Both houses in Column-E are vacant. Each of Column-D and Column-F has at least one occupied house.
4. There is only one house with parking space in Block YY.

Question 89

Which of the following houses is definitely occupied?

- A A1
- B D2
- C B1
- D F2

Handwritten notes:

10 12 ←
0 5 10 ← RAV
← No

Video solution content:

Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX: Columns A, B, C; Rows 1, 2. Houses: A1, B1, C1, A2, B2, C2.

Block YY: Columns D, E, F; Rows 1, 2. Houses: D1, E1, F1, D2, E2, F2.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is also known:

1. The maximum quoted price of a house in Block XX is Rs. 24 lakhs. The minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column-E.
2. Row-1 has two occupied houses, one in each block.
3. Both houses in Column-E are vacant. Each of Column-D and Column-F has at least one occupied house.
4. There is only one house with parking space in Block YY.

VIDEO SOLUTION

Question 90

How many houses are vacant in Block XX?

Row-2: The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX			Block YY		
Column A	Column B	Column C	Column D	Column E	Column F
A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2

Row-1
Row-2

Block

The houses are occupied. The remaining ones are vacant and are the only ones for sale.

The value of a house is the number of its sides adjacent to a road. The values of C2, F2, and B1 are 2, 1, and 0, respectively.

The count of a house is the number of sides of that house adjacent to occupied houses in the same block. For example, E1 and C1 can have the maximum possible neighbour count of 3 and 2, respectively.

The base price of a vacant house is Rs. 10 lakhs if the house does not have any parking space, Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) is calculated as (base price) + 5 * (road adjacency value) + 3 * (neighbour count). The following information is provided:

- The maximum quoted price of a house in Block XX is Rs. 24 lakhs.
- The minimum quoted price of a house in Block YY is Rs. 15 lakhs, and it is in Column-E.
- Row-1 has two occupied houses, one in each block.
- Block houses in Column-F are vacant.

VIDEO SOLUTION

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Question 91

Which of the following options best describes the number of vacant houses in Row-2?

- A Exactly 3
- B Either 3 or 4
- C Exactly 2
- D Either 2 or 3

Row-2: The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX			Block YY		
Column A	Column B	Column C	Column D	Column E	Column F
A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2

Row-1
Row-2

Block

The houses are occupied. The remaining ones are vacant and are the only ones for sale.

The value of a house is the number of its sides adjacent to a road. The values of C2, F2, and B1 are 2, 1, and 0, respectively.

The count of a house is the number of sides of that house adjacent to occupied houses in the same block. For example, E1 and C1 can have the maximum possible neighbour count of 3 and 2, respectively.

The base price of a vacant house is Rs. 10 lakhs if the house does not have any parking space, Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) is calculated as (base price) + 5 * (road adjacency value) + 3 * (neighbour count). The following information is provided:

- The maximum quoted price of a house in Block XX is Rs. 24 lakhs.
- The minimum quoted price of a house in Block YY is Rs. 15 lakhs, and it is in Column-E.
- Row-1 has two occupied houses, one in each block.
- Block houses in Column-F are vacant.

VIDEO SOLUTION

Question 92

What is the maximum possible quoted price (in lakhs of Rs.) for a vacant house in Column-E?

10 12 ←

0 5 10 ← PAV

← No

Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX			Block YY		
Column-A	Column-B	Column-C	Column-D	Column-E	Column-F
A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2

the houses are occupied. The remaining ones are vacant and are the only ones for sale.

Value of a house is the number of its sides adjacent to a road. The values of C2, F2, and B1 are 2, 1, and 0, respectively.

Count of a house is the number of sides of that house adjacent to a road. For example, E1 and C1 can have the maximum possible neighbour count of 3 and 2, respectively.

The base price of a house is Rs. 10 lakhs if the house does not have parking space, 12 lakhs if it does. The quoted price (in lakhs of Rupees) is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is for reference:

- The maximum quoted price of a house in Block XX is Rs. 24 lakhs.
- The minimum quoted price of a house in Block YY is Rs. 15 lakhs, and Column-E.
- Row-1 has two occupied houses, one in each of Block XX and Block YY.
- Row-2 has two occupied houses, one in each of Block XX and Block YY.

VIDEO SOLUTION

Question 93

Which house in Block YY has parking space?

- A E1
- B F2
- C E2
- D F1

10 12 ←

0 5 10 ← PAV

← No

Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX			Block YY		
Column-A	Column-B	Column-C	Column-D	Column-E	Column-F
A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2

the houses are occupied. The remaining ones are vacant and are the only ones for sale.

Value of a house is the number of its sides adjacent to a road. The values of C2, F2, and B1 are 2, 1, and 0, respectively.

Count of a house is the number of sides of that house adjacent to a road. For example, E1 and C1 can have the maximum possible neighbour count of 3 and 2, respectively.

The base price of a house is Rs. 10 lakhs if the house does not have parking space, 12 lakhs if it does. The quoted price (in lakhs of Rupees) is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is for reference:

- The maximum quoted price of a house in Block XX is Rs. 24 lakhs.
- The minimum quoted price of a house in Block YY is Rs. 15 lakhs, and Column-E.
- Row-1 has two occupied houses, one in each of Block XX and Block YY.
- Row-2 has two occupied houses, one in each of Block XX and Block YY.

VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile



cracku

Instructions

Fuel contamination levels at each of 20 petrol pumps P1, P2, ..., P20 were recorded as either high, medium, or low.

- Contamination levels at three pumps among P1 - P5 were recorded as high.
- P6 was the only pump among P1 - P10 where the contamination level was recorded as low.
- P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

4. High contamination levels were not recorded at any of the pumps P16 - P20.
5. The number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded.

Question 94

What best can be said about the number of pumps at which the contamination levels were recorded as medium?

- A At least 8
- B More than 4
- C Exactly 8
- D At most 9

[VIDEO SOLUTION](#)

Question 95

If contamination level at P15 was recorded as medium, then which of the following MUST be FALSE?

- A Contamination levels at P13 and P17 were recorded as the same.
- B Contamination levels at P11 and P16 were recorded as the same.
- C Contamination level at P14 was recorded to be higher than that at P15.
- D Contamination levels at P10 and P14 were recorded as the same.

[VIDEO SOLUTION](#)

Question 96

Which of the following **MUST** be true?

- A The contamination level at P20 was recorded as medium.
- B The contamination level at P13 was recorded as low.
- C The contamination level at P12 was recorded as high.
- D The contamination level at P10 was recorded as high.

Fuel contamination levels at each of 20 petrol pumps P1, P2, ..., P20 were recorded as either high, medium, or low.

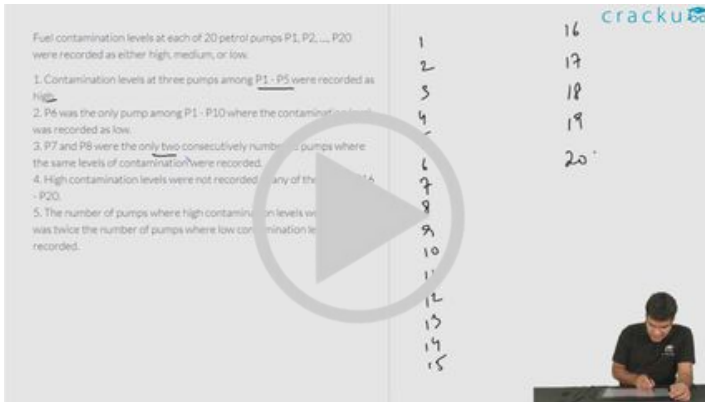
1. Contamination levels at three pumps among P1 - P5 were recorded as high.

2. P6 was the only pump among P1 - P10 where the contamination level was recorded as low.

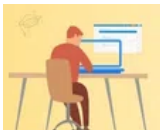
3. P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

4. High contamination levels were not recorded for any of the pumps P14 - P20.

5. The number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded.



 VIDEO SOLUTION





CAT Sectional Tests



Take Here

Question 97

If the contamination level at P11 was recorded as low, then which of the following **MUST** be true?

- A The contamination level at P12 was recorded as high.
- B The contamination level at P15 was recorded as medium.
- C The contamination level at P18 was recorded as low.
- D The contamination level at P14 was recorded as medium.

Fuel contamination levels at each of 20 petrol pumps P1, P2, ..., P20 were recorded as either high, medium, or low.

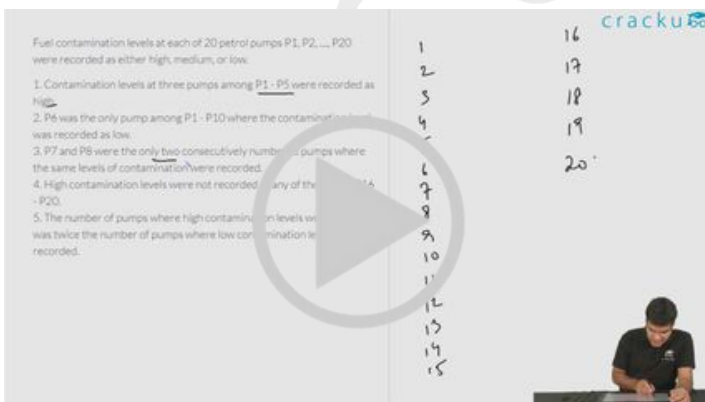
1. Contamination levels at three pumps among P1 - P5 were recorded as high.

2. P6 was the only pump among P1 - P10 where the contamination level was recorded as low.

3. P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

4. High contamination levels were not recorded for any of the pumps P14 - P20.

5. The number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded.



 VIDEO SOLUTION

Instructions

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared consecutively in MT as well as ET.
- Chetan prepared the third question in both MT and ET; and Esha prepared the eighth question in both.
- Fakir prepared the first question of MT and the last one in ET. Dave prepared the last question of MT and the first one in ET.

Question 98

The second question in ET was prepared by:

- A Chetan
- B Beti
- C Esha
- D Dave

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared consecutively in MT as well as ET.
- Chetan prepared the third question in both MT and ET; and Esha prepared the eighth question in both.
- Fakir prepared the first question of MT and the last one in ET. Dave prepared the last question of MT and the first one in ET.

	A	B	C	D	E	F
MT	5	10	15	2		
ET	3	3	2			

VIDEO SOLUTION

Question 99

How many 5-mark questions were there in MT and ET combined?

- A 13
- B 12
- C 10
- D Cannot be determined

cracku

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared

Handwritten notes on a whiteboard:

BTMT
 A B C D E F
 MT
 ET
 5 10 15
 3 3 2



VIDEO SOLUTION

cracku

CAT Study Material

All CAT Free Resources



Question 100

Who prepared 15-mark questions for MT and ET?

- A Only Beti, Dave, Esha and Fakir
- B Only Dave and Fakir
- C Only Esha and Fakir
- D Only Dave, Esha and Fakir

cracku

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared

Handwritten notes on a whiteboard:

BTMT
 A B C D E F
 MT
 ET
 5 10 15
 3 3 2



VIDEO SOLUTION

Question 101

Which of the following questions did Beti prepare in ET?

- A Seventh question
- B Fourth question
- C Ninth question
- D Tenth question

cracku

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared

Handwritten notes on a whiteboard:

BTMT
 A B C D E F
 MT
 ET
 5 10 15
 3 3 2

VIDEO SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

- The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.
- Three persons, including the one who eats four vadas eat without chutney.
- Sandeep does not take any chutney.
- The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.
- Daljit eats idli with chutney and also eats vada.
- Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.
- Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Question 102

Which of the following statements is true?

- Mukesh eats 8 idlis and 4 vadas but no chutney.
- The person who eats 5 idlis and 1 vada does not take chutney.
- The person who eats equal number of vadas and idlis also takes chutney.
- The person who eats 4 idlis and 2 vadas also takes chutney.

cracku

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

i. The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.

ii. Three persons, including the one who eats four vadas eat without chutney.

iii. Sandeep does not take any chutney.

iv. The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.

v. Daljit eats idli with chutney and also eats vada.

vi. Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.

vii. Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Which one of the following statements is true?

A) Daljit eats 5 idlis
C) Bimal eats 1 idli.

	Idli	Vadas	Chutney
Ignesh			

Handwritten notes: 4, a, 3a, a=0

VIDEO SOLUTION



Question 103

Which one of the following statements is true?

- A Daljit eats 5 idlis
- B Ignesh eats 8 idlis
- C Bimal eats 1 idli.
- D Bimal eats 6 idlis.

cracku

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

i. The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.

ii. Three persons, including the one who eats four vadas eat without chutney.

iii. Sandeep does not take any chutney.

iv. The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.

v. Daljit eats idli with chutney and also eats vada.

vi. Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.

vii. Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Which one of the following statements is true?

A) Daljit eats 5 idlis
C) Bimal eats 1 idli.

	Idli	Vadas	Chutney
Ignesh			

Handwritten notes: 4, a, 3a, a=0

VIDEO SOLUTION

Question 104

Which of the following statements is true?

- A Sandeep eats 2 vadas.
- B Mukesh eats 4 vadas.
- C Ignesh eats 6 vadas.
- D Bimal eats 2 vadas.

cracku

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

i. The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.

ii. Three persons, including the one who eats four vadas eat without chutney.

iii. Sandeep does not take any chutney.

iv. The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.

v. Daljit eats idli with chutney and also eats vada.

vi. Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.

vii. Binal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Binal.

Which one of the following statements is true?

A) Daljit eats 5 idlis
C) Binal eats 1 idli

B) Ignesh eats 8 idlis
D) Binal eats 6 idlis

	Idli	Vadas	Chutney
Ignesh			

VIDEO SOLUTION

Instructions

DIRECTIONS for the following two questions: Answer the questions on the basis of the information given below.

Some children were taking free throws at the basketball court in school during lunch break. Below are some facts about how many baskets these children shot.

- Ganesh shot 8 baskets less than Ashish.
- Dhanraj and Ramesh together shot 37 baskets.
- Jugraj shot 8 baskets more than Dhanraj.
- Ashish shot 5 baskets more than Dhanraj.
- Ashish and Ganesh together shot 40 baskets.

Question 105

Which of the following statements is true?

- Ramesh shot 18 baskets and Dhanraj shot 19 baskets
- Ganesh shot 24 baskets and Ashish shot 16 baskets
- Jugraj shot 19 baskets and Dhanraj shot 27 baskets
- Dhanraj shot 11 baskets and Ashish shot 16 baskets

VIEW SOLUTION

100+ CAT Quant Questions



Question 106

Which of the following statements is true?

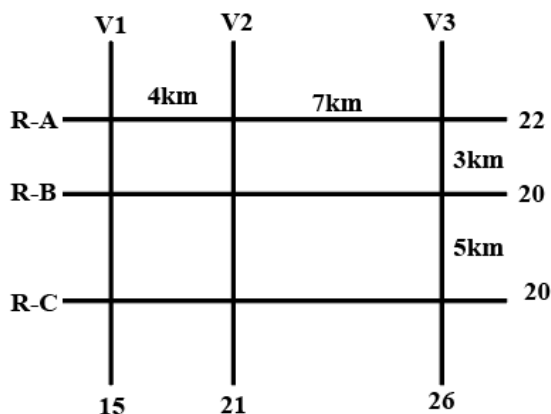
- Dhanraj and Jugraj together shot 46 baskets.
- Ganesh shot 18 baskets and Ramesh shot 21 baskets.
- Dhanraj shot 3 more baskets than Ramesh.
- Ramesh and Jugraj together shot 29 baskets.

Instructions

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km.



Question 107

How many ATMs have cash requirements of Rs. 10 Lakhs or more?

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km.

Question 108

Which of the following statements is correct?

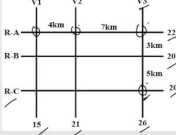
- A There is no ATM placed at the (R-C, V2) intersection.
- B The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 9 Lakhs.
- C The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 8 Lakhs.

D The cash requirement of the ATM placed at the (R-C, V2) intersection cannot be uniquely determined.

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R and V3 is 12 km.



VIDEO SOLUTION

100+ CAT DILR Questions



Question 109

Which of the following two statements is/are DEFINITELY true?

Statement A: Each of R-A, R-B, and R-C has two ATMs.

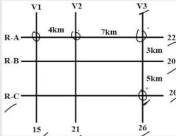
Statement B: Each of V1, V2, and V3 has two ATMs.

- A Only Statement A
- B Neither Statement A nor Statement B
- C Only Statement B
- D Both Statement A and Statement B

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R and V3 is 12 km.



VIDEO SOLUTION

Question 110

What best can be said about the road distance (in km) between the ATMs having the second highest and the second lowest cash requirements?

- A 4 km

- B 7 km
- C 5 km
- D Either 4 km or 7 km

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-B and V3 is 12 km.

VIDEO SOLUTION

Question 111

What is the number of ATMs whose locations and cash requirements can both be uniquely determined?

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-B and V3 is 12 km.

VIDEO SOLUTION

100+ CAT VARC Questions



Instructions

A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given a number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter V.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question 112

Suppose eight cars have arrived, of which two have left. Also suppose that car 4 is a compact and car 7 is an SUV. Which of the following is a POSSIBLE current configuration of the parking lot?

- A 8, 2, 3, V, 6, 5, 7
- B V, 2, 3, 7, 5, 6, 8
- C 8, 2, 3, V, 5, 7, 6
- D 8, 2, 3, V, 5, 6, 7

cracku

A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter V.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question (1)

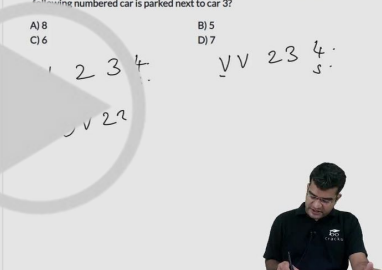
Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

A) 8

C) 6

B) 5

D) 7



VIDEO SOLUTION

Question 113

Suppose the sequence at some point of time is 4, 5, 6, V, 3. Which of the following is NOT necessarily true?

- A Car 4 is a compact.
- B Car 1 is an SUV.
- C Car 3 is an SUV
- D Car 5 is a compact.

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A) 8
B) 5
C) 6
D) 7

VIDEO SOLUTION

Question 114

Suppose that car 4 is not the first car to leave and that the sequence at a time between the arrival of the car 7 and car 8 is V, 7, 3, 6, 5. Then which of the following statements MUST be false?

- A Car 2 is a compact.
- B Car 7 is a compact.
- C Car 4 is an SUV.
- D Car 6 is a compact.

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A) 8
B) 5
C) 6
D) 7

VIDEO SOLUTION

CAT Expected Questions PDF



Question 115

Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

- A 8

B 5

C 6

D 7

cracku

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A) 8
B) 5
C) 6
D) 7

Handwritten notes: 1 2 3 4, V V 2 3 4, V V 2 3 4



VIDEO SOLUTION

Instructions

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

Question 116

What is the minimum number of Blue beads in any configuration?

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Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

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Every unique, complete arrangement of twenty five beads is called a configuration.

G	B			
B				
	B	G	B	
B	G	B	B	



VIDEO SOLUTION

Question 117

Two Red beads have been placed in 'second row, third column' and 'third row, second column'. How many more Red beads can be placed so as to maximise the number of Red beads used in the configuration?

cracku

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G ₁	B ₁			
B ₁				
	B	G	B	
B	G	B	B	

VIDEO SOLUTION

CAT Important Topics

Question 118

What is the maximum possible number of Red beads that can appear in any configuration?

cracku

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1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G ₁	B ₁			
B ₁				
	B	G	B	
B	G	B	B	

VIDEO SOLUTION

Question 119

The total number of possible configurations using beads of only two colours is:

cracku

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G ₁	B ₁			
B ₁				
	B	G	B	
B	G	B	B	

 VIDEO SOLUTION

Instructions

Direction for the following two questions: Answer the questions based on the following information. Amar, Akbar and Anthony are three friends. Only three colours are available for their shirts, viz. red, green and blue. Amar does not wear red shirt. Akbar does not wear green shirt. Anthony does not wear blue shirt.

Question 120

If two of them wear the same colour, then how many of the following must be false?

- I. Amar wears blue and Akbar does not wear green
- II. Amar does not wear blue and Akbar wears blue
- III. Amar does not wear blue and Akbar does not wear blue
- IV. Amar wears green, Akbar does not wear red, Anthony does not wear green

- A None
- B One
- C Two
- D Three

 VIEW SOLUTION



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Question 121

If Akbar and Anthony wear the same colour of shirts, then which of the following is not true?

- A Amar wears blue and Akbar wears green
- B Amar wears green and Akbar wears red
- C Amar wears blue and Akbar does not wear blue
- D Anthony wears red

 VIEW SOLUTION